

BIOS 755: Multilevel Logistic Regression

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Introduction

- ▶ Multilevel modeling can be applied to logistic regression and other generalized linear models
- ▶ This will be similar to the linear case where coefficients and random effects will be grouped at different levels in the data.
- ▶ The only difference is how level 1 error, i.e., the residual, is captured. This latter point is of interest when we want to calculate the ICC.

Examples of GLMM

- ▶ Binary logistic model with random intercepts:

$$\text{logit}\{P(Y_{ij} = 1|b_j)\} = \beta_0 + \beta_1 X_i + b_j$$

$$\log \left\{ \frac{P(Y_{ij} = 1|b_j)}{1 - P(Y_{ij} = 1|b_j)} \right\} = \beta_0 + \beta_1 X_j + b_j$$

$$P(Y_{ij} = 1|b_j) = \frac{e^{\beta_0 + \beta_1 X_j + b_j}}{1 + e^{\beta_0 + \beta_1 X_j + b_j}}$$

with $b_j \sim N(0, \sigma^2)$.

- ▶ Here, e^{β_1} gives the level 1 change in the odds for a 1-unit increase in X .

Examples of GLMM

- ▶ Random coefficients Poisson regression model:

$$\log\{E(Y_{ij}|\mathbf{b}_j)\} = \log(n_{ij}) + \beta_0 + \beta_1 G_i + \beta_2 X_{ij} + \beta_3 G_i X_{ij} + b_{j0} + b_{j1} X_{ij}$$

$$\log\left\{E\left(\frac{Y_{ij}}{n_{ij}}\middle|\mathbf{b}_j\right)\right\} = \beta_0 + \beta_1 G_i + \beta_2 X_{ij} + \beta_3 G_i X_{ij} + b_{j0} + b_{j1} X_{ij}$$

$$E\left(\frac{Y_{ij}}{n_{ij}}\middle|\mathbf{b}_j\right) = \exp(\beta_0 + \beta_1 G_i + \beta_2 X_{ij} + \beta_3 G_i X_{ij} + b_{j0} + b_{j1} X_{ij})$$

where $\mathbf{b}_j = (b_{j0} \ b_{j1})$ with a random intercept and a random effect of X and $\mathbf{b}_j \sim N(0, \mathbf{G})$.

- ▶ Here, e^{β_1} gives the level 1 change in the rate (i.e., $E(Y_{ij})$ per each n_{ij}) between treatment and control groups.

Example

Guatemalan immunization campaign

- ▶ Data are available from the National Survey of Maternal and Child Health conducted in Guatemala in 1987
- ▶ A nationally representative sample of 5160 women aged between 15 and 44 were interviewed
- ▶ The questionnaire included questions determining the immunization status of children who were born in the previous 5 years and alive at the time of the interview

Example

- ▶ Beginning in 1986, the Guatemalan government undertook several campaigns to immunize the population against major childhood diseases.
- ▶ An important explanatory variable is whether the child was at least 2 years old at the time of the interview, in which case the child was old enough to be immunized during the 1986 campaign.
- ▶ If this variable is associated with immunization, there is some indication that the government campaign worked.

What are the levels of data?

Two-level model

- ▶ As we discussed last week, a two-level model would be similar to what we've done before

$$\text{logit}\{P(Y_{ij} = 1|\mathbf{X})\} = \beta_0 + \beta_1 X_{ij} + b_{j0}$$

where

- ▶ Y_{ij} is the immunization status for the i th child from the j th mother
 - ▶ X_{ij} indicates that the child is at least 2 years old.
 - ▶ $b_{j0} \sim N(0, \sigma_b)$ is the mother level random intercept
- ▶ **Question:** how do we estimate the ICC?

Estimating the ICC (two-level)



- ▶ To estimate the ICC, we need to put the logistic model into a latent variable formulation.
- ▶ In this model, we assume underlying the observed dichotomous response (whether the child was immunized), there is an unobserved or latent continuous response.
- ▶ This latent response represents the propensity to be immunized.
- ▶ If this latent response is greater than zero, then the observed response is 1, if not, the response is 0.

Estimating the ICC (two-level)

- ▶ The latent variable formulation is

$$Y_{ij}^* = \beta_0 + \beta_1 X_{ij} + b_{j0} + \varepsilon_{ij}$$

where

$$Y_{ij}^* > 0 \rightarrow Y_{ij} = 1$$

$$Y_{ij}^* \leq 0 \rightarrow Y_{ij} = 0$$

and $E(\varepsilon_{ij}) = 0$

Estimating the ICC (two-level)

- ▶ In logistic regression the error ε_{ij} is assumed to have a logistic distribution where

$$\Pr(\varepsilon_{ij} < \tau | \mathbf{X}_{ij}) = \frac{\exp(\tau)}{1 + \exp(\tau)}$$

and $\text{Var}[\varepsilon_i | x_i] = \frac{\pi^2}{3} \approx 3.29$

- ▶ The estimate the ICC for a two-level model is

$$\text{ICC} = \frac{\sigma_b^2}{\sigma_b^2 + \frac{\pi^2}{3}} \approx \frac{\sigma_b^2}{\sigma_b^2 + 3.29}$$



Median Odds Ratio

- ▶ Similar to the ICC, the Median Odds Ratio (MOR) quantifies the amount of clustering for logistic regression.
- ▶ The goal of the MOR is to give the amount of clustering using the scale of ORs.
- ▶ Suppose two kids have equal predictor variables but are from different mothers. The OR of immunization between these kids is:

$$OR_{jk} = \frac{P(Y_{ij} = 1)}{P(Y_{i'k} = 1)} = e^{b_j - b_k} \quad (1)$$

since the random effects are the only difference.

- ▶ The idea of MOR is to calculate all the possible ORs obtained from (1) for all j and k such that $b_j > b_k$.

Median Odds Ratio

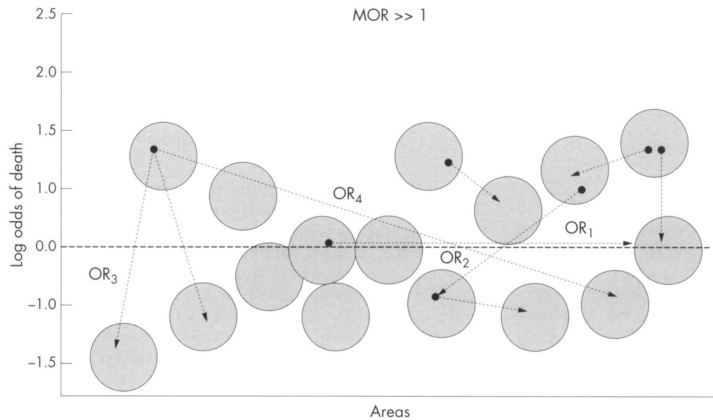


Figure from: Merlo, J., et al. (2006). A brief conceptual tutorial of multilevel analysis in social epidemiology: using measures of clustering in multilevel logistic regression to investigate contextual phenomena. *Journal of Epidemiology & Community Health*, 60(4), 290-297.

Median Odds Ratio

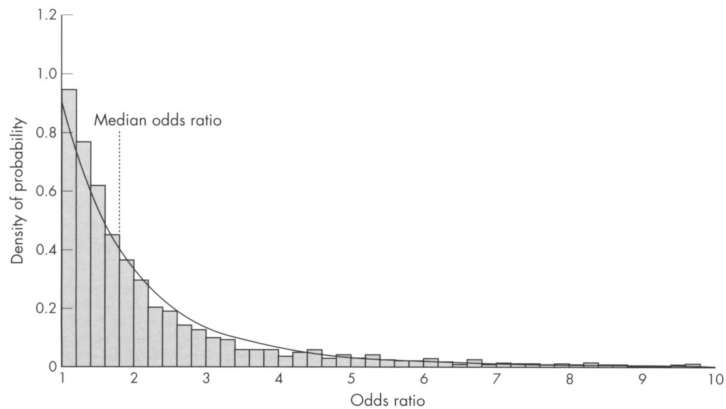


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Median Odds Ratio

- ▶ Calculating the MOR is straightforward:

$$MOR = \exp\left(0.954\sqrt{\sigma_b^2}\right) = \exp(0.954\sigma_b)$$

- ▶ If $MOR = 1$, then there would be no differences between mothers in the probability of being immunized.
- ▶ If $MOR = 1.8$ then the median difference between mothers increased the child level odds of being immunized by 80% when randomly picking out two mothers.
- ▶ That is, if a child was to randomly change to a different mother that had higher immunization probability, the median increase in the child's odds of immunization is by a factor of 1.8.
 - ▶ So, 50% of the time it would be lower than 1.8, 50% of the time it would be higher than 1.8.

Three-level model

- ▶ In our example, there is actually a third level (community).
- ▶ It is of interest to fit the model

$$\text{logit}\{P(Y_{ijk} = 1|\mathbf{X})\} = \beta_0 + \beta_1 X_{ijk} + b_{jk0} + b_{k0}$$

where

- ▶ Y_{ijk} is the immunization status for the i th child from the j th mother in the k th community
- ▶ X_{ijk} is the indicator that the child is at least 2 years old
- ▶ $b_{jk0} \sim N(0, \sigma_{(2)}^2)$ is the mother-level random intercept
- ▶ $b_{k0} \sim N(0, \sigma_{(3)}^2)$ is the community-level random intercept

Estimating the ICC (three-level)

- Correlation across mothers within the same community

$$\rho(comm) = \text{corr}(Y_{ijk}^*, Y_{i'j'k}^*) = \frac{\sigma_3^2}{\sigma_2^2 + \sigma_3^2 + \pi^2/3}$$

- Correlation across children for the same mother (ignoring community) is

$$\rho(mother) = \text{corr}(Y_{ijk}^*, Y_{i'jk}^*) = \frac{\sigma_2^2}{\sigma_2^2 + \sigma_3^2 + \pi^2/3}$$

- Correlation across children for the same mother and within the same community

$$\rho(mother, comm) = \text{corr}(Y_{ijk}^*, Y_{i'jk}^*) = \frac{\sigma_2^2 + \sigma_3^2}{\sigma_2^2 + \sigma_3^2 + \pi^2/3}$$

Estimating the MOR (three-level)

- ▶ MOR for changing mothers and staying in the same community

$$MOR(comm) = \exp(0.954\sigma_3)$$

- ▶ MOR for changing communities and mother staying with the same

$$MOR(mother) = \exp(0.954\sigma_2)$$